

Mamak Order Tracker

Web Storage – localStorage Project

JOB SHEET 12 – WEB STORAGE (localStorage)

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Project: Mamak Order Tracker (self-made project)

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Programme: Diploma in Computer Technology (Big Data), TVETMARA Lumut

Files: haziq-12-mamak-order-tracker-web-storage.html, .css, .js

1. Project Description

Mamak Order Tracker is a small order-taking web app for a mamak stall. The waiter taps menu items to build an order, chooses a table and saves the order. Everything is stored in the browser using `localStorage`, so the current order, order history, sales summary, theme and selected table are still there after refreshing or closing the browser.

2. Main Features

- Menu with 8 items and prices in RM; tap to add to the current order.
- Current order with quantity buttons (+ / -), automatic total and table selection (Table 1–10 or Takeaway).
- Place Order saves the order with an order number, time, table, items and total.
- Order history with filter by table and delete button.
- Today's summary: orders today, sales today, best seller today, all-time sales.
- Storage Inspector: lists every `localStorage` key used by the app and its size.
- Export JSON, Reset All Data, and light/dark theme saved between visits.
- Two open tabs stay in sync using the `storage` event.

3. Screenshots

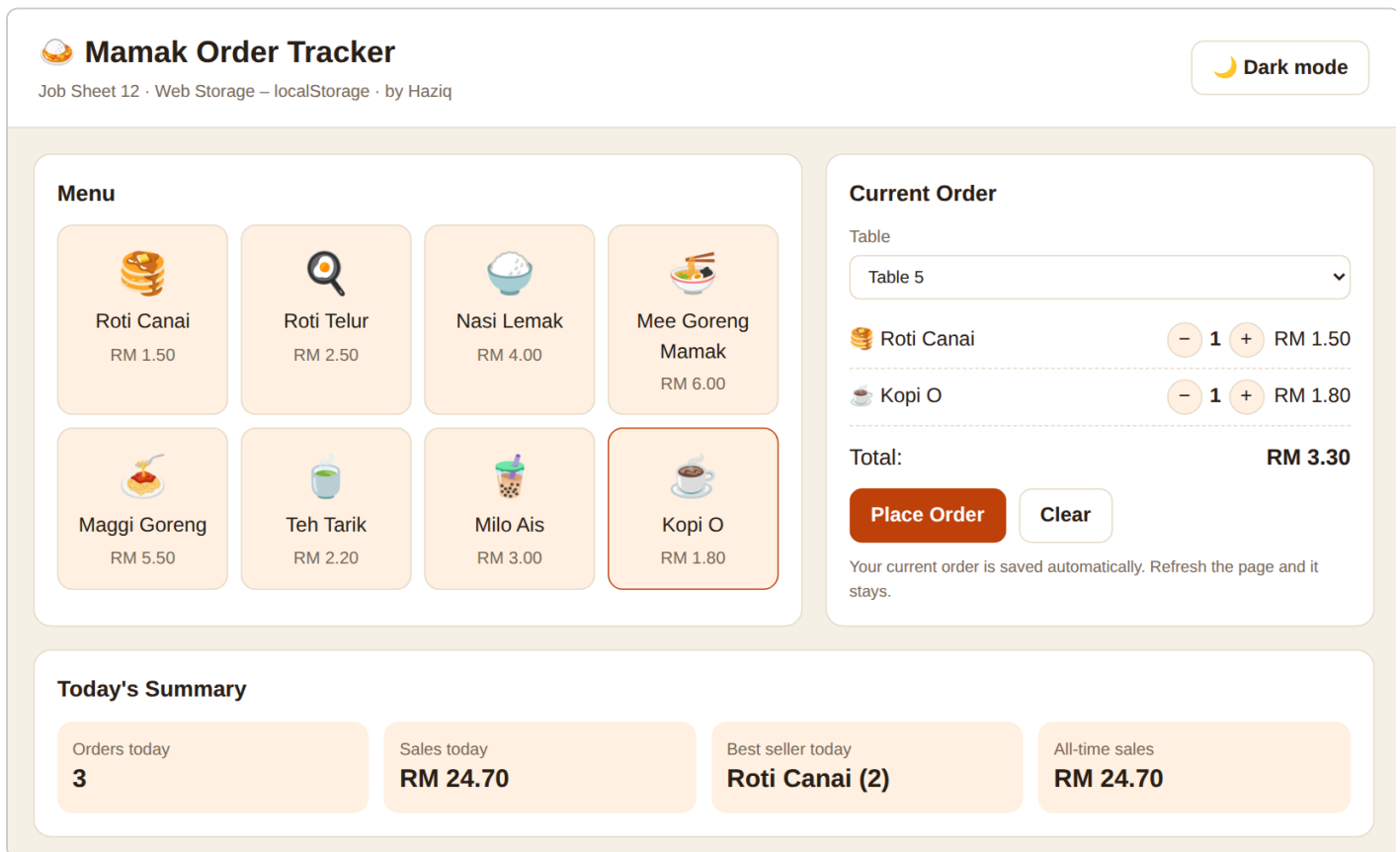


Figure 1 – Main screen with menu, current order (Table 5) and today's summary.

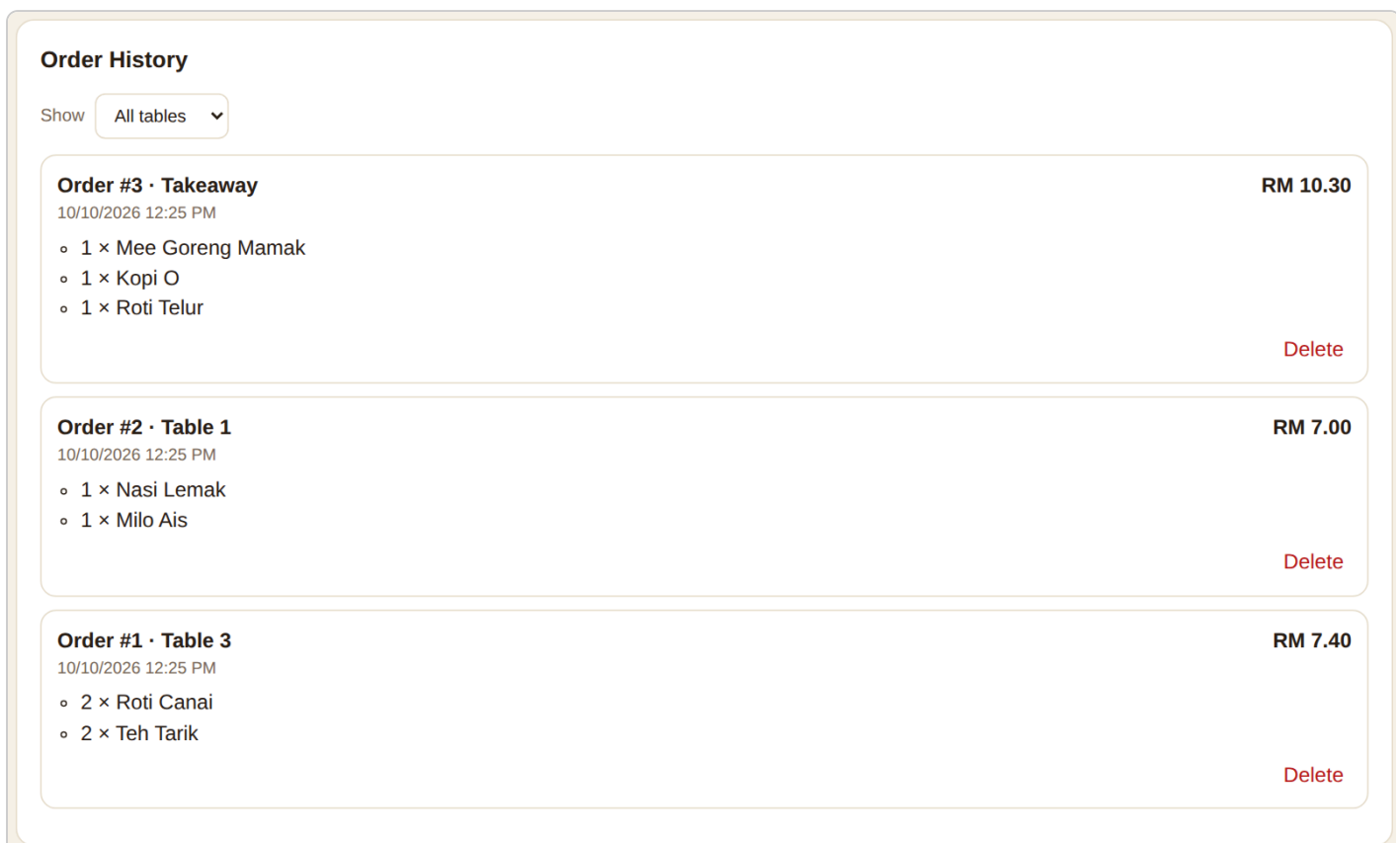


Figure 2 – Order history showing saved orders.

Storage Inspector

Live view of what this app keeps in `localStorage`.

Key	Stores	Size (chars)
<code>haziq12_mamak_cart</code>	Current order (item id → quantity)	33
<code>haziq12_mamak_counter</code>	Last order number used	22
<code>haziq12_mamak_orders</code>	Saved order history	635
<code>haziq12_mamak_table</code>	Selected table	28

Export JSON

Reset All Data

Figure 3 – Storage Inspector showing the localStorage keys and their sizes.

Mamak Order Tracker

Job Sheet 12 · Web Storage – localStorage · by Haziq

Light mode

Menu

Roti Canai

RM 1.50

Roti Telur

RM 2.50

Nasi Lemak

RM 4.00

Mee Goreng Mamak

RM 6.00

Maggi Goreng

RM 5.50

Teh Tarik

RM 2.20

Milo Ais

RM 3.00

Kopi O

RM 1.80

Current Order

Table

Table 5

Roti Canai

–

1

+

RM 1.50

Kopi O

–

1

+

RM 1.80

Total:

RM 3.30

Place Order

Clear

Your current order is saved automatically. Refresh the page and it stays.

Today's Summary

Orders today

3

Sales today

RM 24.70

Best seller today

Roti Canai (2)

All-time sales

RM 24.70

Figure 4 – Dark mode (the choice is saved in localStorage).

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4. How localStorage Is Used

All keys start with the prefix `haziq12_mamak_` so they do not clash with other job sheets in the same browser.

Key	Stores	Example value
<code>haziq12_mamak_cart</code>	Current order (item id → quantity)	<code>{"rc":1,"ko":1}</code>
<code>haziq12_mamak_orders</code>	Saved order history (array of orders)	<code>[{"no":3,"table":"Takeaway","items":[...],"total":1230}, ...]</code>
<code>haziq12_mamak_counter</code>	Last order number used	<code>3</code>
<code>haziq12_mamak_theme</code>	Light or dark theme	<code>"dark"</code>
<code>haziq12_mamak_table</code>	Selected table	<code>"Table 5"</code>

Web Storage methods and techniques used

Technique	Where it is used
<code>localStorage.setItem()</code>	Saving cart, orders, counter, theme and table
<code>localStorage.getItem()</code>	Loading saved data when the page opens
<code>localStorage.removeItem()</code>	Emptying the cart and Reset All Data
<code>localStorage.length</code> and <code>key(i)</code>	Storage Inspector lists all keys with the app prefix
<code>JSON.stringify()</code> / <code>JSON.parse()</code>	Storing objects and arrays as text and reading them back
<code>try / catch</code>	Handles blocked or full storage and corrupted data
<code>storage event</code>	Updates one tab when another tab changes the data

5. Testing

The app was tested in a real browser (Chromium). Results:

- Added 2 Roti Canai and 1 Teh Tarik, refreshed the page: the current order (RM 5.20) was still there.
- Placed the order for Table 4: order saved, cart emptied, history and summary updated.
- Switched to dark mode and refreshed: dark mode and the last selected table were remembered.
- Filtering history by table, deleting one order and Reset All Data all worked.
- Changing the cart in one tab updated the other tab through the storage event.
- No JavaScript errors were reported in the console.

6. How to Run

Put the three files in the same folder and open `haziq-12-mamak-order-tracker-web-storage.html` in a browser. No server or Internet connection is needed.

7. Source Code – haziq-12-mamak-order-tracker-web-storage.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Mamak Order Tracker – Web Storage (localStorage)</title>
<link rel="stylesheet" href="haziq-12-mamak-order-tracker-web-storage.css">
</head>
<body>
<header class="top">
  <div>
    <h1>🍲 Mamak Order Tracker</h1>
    <p>Job Sheet 12 · Web Storage – localStorage · by Haziq</p>
  </div>
  <button id="themeBtn" class="ghost" type="button">🌙 Dark mode</button>
</header>

<main class="layout">
  <section class="card" aria-labelledby="menuTitle">
    <h2 id="menuTitle">Menu</h2>
    <div id="menu" class="menu"></div>
  </section>

  <section class="card" aria-labelledby="orderTitle">
    <h2 id="orderTitle">Current Order</h2>
    <label class="field">Table
      <select id="tableSel"></select>
    </label>
    <p id="cartEmpty" class="empty">No items yet. Tap a menu item to add it.</p>
    <ul id="cartList" class="cart"></ul>
    <div class="total">Total: <strong id="cartTotal">RM 0.00</strong></div>
    <div class="row">
      <button id="placeBtn" class="primary" type="button">Place Order</button>
      <button id="clearCartBtn" class="ghost" type="button">Clear</button>
    </div>
    <p class="hint">Your current order is saved automatically. Refresh the page and it stays.</p>
  </section>

  <section class="card wide" aria-labelledby="statsTitle">
    <h2 id="statsTitle">Today's Summary</h2>
    <div id="stats" class="stats"></div>
  </section>

  <section class="card wide" aria-labelledby="histTitle">
    <h2 id="histTitle">Order History</h2>
    <label class="field inline">Show
      <select id="filterTable"></select>
    </label>
    <p id="histEmpty" class="empty">No orders saved yet.</p>
    <ul id="history" class="history"></ul>
  </section>

  <section class="card wide" aria-labelledby="storeTitle">
    <h2 id="storeTitle">Storage Inspector</h2>
    <p class="hint">Live view of what this app keeps in <code>localStorage</code>.</p>
    <div class="scroll">
      <table id="storageTable">
        <thead><tr><th>Key</th><th>Stores</th><th>Size (chars)</th></tr></thead>
        <tbody></tbody>
      </table>
    </div>
    <div class="row">
      <button id="exportBtn" class="ghost" type="button">Export JSON</button>
      <button id="resetBtn" class="danger" type="button">Reset All Data</button>
    </div>
  </section>
</main>

<p id="toast" class="toast" role="status" aria-live="polite"></p>
<script src="haziq-12-mamak-order-tracker-web-storage.js"></script>
</body>
</html>
```

8. Source Code – haziq-12-mamak-order-tracker-web-storage.css

```
:root{
  --bg:#f6f1e7;--card:#ffffff;--text:#2b2118;--muted:#7a6a5a;--line:#e6dcccb;
  --accent:#c2410c;--accent-text:#ffffff;--danger:#b91c1c;--ok:#15803d;--chip:#fff4e5;
}
:root[data-theme="dark"]{
  --bg:#17130f;--card:#241d17;--text:#f3e9da;--muted:#b3a38f;--line:#3a2f25;
  --accent:#f59e0b;--accent-text:#1b1209;--danger:#f87171;--ok:#4ade80;--chip:#33281c;
}
*{box-sizing:border-box}
body{margin:0;font-family:"Segoe UI",Arial,sans-serif;background:var(--bg);color:var(--text);line-height:1.45}
.top{display:flex;justify-content:space-between;align-items:center;gap:12px;padding:18px 24px;border-bottom:1px
solid var(--line);background:var(--card)}
.top h1{margin:0;font-size:1.5rem}
.top p{margin:2px 0 0;color:var(--muted);font-size:.85rem}
.layout{max-width:1100px;margin:0 auto;padding:20px;display:grid;grid-template-columns:1.4fr 1fr;gap:18px}
.card{background:var(--card);border:1px solid var(--line);border-radius:14px;padding:18px}
.card.wide{grid-column:1 / -1}
.card h2{margin:0 0 12px;font-size:1.1rem}
.menu{display:grid;grid-template-columns:repeat(auto-fill,minmax(130px,1fr));gap:10px}
.menu button{display:flex;flex-direction:column;align-items:center;gap:4px;padding:14px 8px;border:1px solid var(--
line);border-radius:12px;background:var(--chip);color:var(--text);cursor:pointer;font:inherit}
.menu button:hover{border-color:var(--accent)}
.menu .icon{font-size:2rem}
.menu .price{color:var(--muted);font-size:.85rem}
.field{display:flex;flex-direction:column;gap:4px;font-size:.85rem;color:var(--muted);margin-bottom:10px}
.field.inline{flex-direction:row;align-items:center;gap:8px;margin-bottom:12px}
select{padding:8px 10px;border:1px solid var(--line);border-radius:8px;background:var(--card);color:var(--
text);font:inherit}
.cart,.history{list-style:none;margin:0;padding:0}
.cart li{display:flex;align-items:center;gap:8px;padding:8px 0;border-bottom:1px dashed var(--line)}
.cart .name{flex:1}
.cart .qty{display:flex;align-items:center;gap:6px}
.qty button{width:28px;height:28px;border-radius:50%;border:1px solid var(--line);background:var(--
chip);color:var(--text);cursor:pointer}
.total{display:flex;justify-content:space-between;margin:12px 0;font-size:1.1rem}
.row{display:flex;gap:10px;flex-wrap:wrap}
button{font:inherit}
.primary,.ghost,.danger{padding:9px 16px;border-radius:10px;border:1px solid transparent;cursor:pointer;font-
weight:600}
.primary{background:var(--accent);color:var(--accent-text)}
.ghost{background:transparent;color:var(--text);border-color:var(--line)}
.danger{background:transparent;color:var(--danger);border-color:var(--danger)}
.empty{color:var(--muted);font-style:italic;margin:6px 0}
.hint{color:var(--muted);font-size:.8rem;margin:10px 0 0}
.stats{display:grid;grid-template-columns:repeat(auto-fit,minmax(160px,1fr));gap:12px}
.stat{background:var(--chip);border-radius:12px;padding:12px}
.stat small{display:block;color:var(--muted)}
.stat strong{font-size:1.25rem}
.history > li{border:1px solid var(--line);border-radius:12px;padding:12px;margin-bottom:10px}
.history li.head{display:flex;justify-content:space-between;gap:8px;flex-wrap:wrap;font-weight:600}
.history .meta{color:var(--muted);font-size:.8rem}
.history ul{margin:6px 0 8px;padding-left:18px}
.history .foot{display:flex;justify-content:space-between;align-items:center}
.history .del{background:transparent;border:0;color:var(--danger);cursor:pointer}
.scroll{overflow-x:auto;margin-bottom:12px}
table{border-collapse:collapse;width:100%;font-size:.9rem}
th,td{border:1px solid var(--line);padding:8px 10px;text-align:left}
th{background:var(--chip)}
code{background:var(--chip);padding:1px 5px;border-radius:5px}
.toast{position:fixed;left:50%;bottom:22px;transform:translateX(-50%) translateY(20px);background:var(--
text);color:var(--bg);padding:10px 18px;border-radius:999px;opacity:0;pointer-events:none;transition:.25s}
.toast.show{opacity:1;transform:translateX(-50%) translateY(0)}
@media (max-width:760px){.layout{grid-template-columns:1fr;padding:12px}.top{padding:14px 16px}}
```

9. Source Code – haziq-12-mamak-order-tracker-web-storage.js

```
"use strict";

/* ----- Config ----- */
const PREFIX = "haziq12_mamak_";
const KEYS = {
  cart: PREFIX + "cart",
  orders: PREFIX + "orders",
  counter: PREFIX + "counter",
  theme: PREFIX + "theme",
  table: PREFIX + "table"
};
const KEY_INFO = {
  [KEYS.cart]: "Current order (item id → quantity)",
  [KEYS.orders]: "Saved order history",
  [KEYS.counter]: "Last order number used",
  [KEYS.theme]: "Light or dark theme",
  [KEYS.table]: "Selected table"
};
const MENU = [
  { id: "rc", name: "Roti Canai", price: 150, icon: "🥞" },
  { id: "rt", name: "Roti Telur", price: 250, icon: "🥞" },
  { id: "nl", name: "Nasi Lemak", price: 400, icon: "🍛" },
  { id: "mg", name: "Mee Goreng Mamak", price: 600, icon: "🍜" },
  { id: "mgi", name: "Maggi Goreng", price: 550, icon: "🍜" },
  { id: "tt", name: "Teh Tarik", price: 220, icon: "🥤" },
  { id: "mi", name: "Milo Ais", price: 300, icon: "🥤" },
  { id: "ko", name: "Kopi O", price: 180, icon: "☕" }
];
const TABLES = Array.from({ length: 10 }, (_, i) => "Table " + (i + 1)).concat("Takeaway");

/* ----- Storage helpers (all wrapped in try/catch) ----- */
const store = {
  get(key, fallback) {
    try {
      const raw = localStorage.getItem(key);
      return raw === null ? fallback : JSON.parse(raw);
    } catch (e) { return fallback; }
  },
  set(key, value) {
    try { localStorage.setItem(key, JSON.stringify(value)); return true; }
    catch (e) { toast("Could not save: storage is full or blocked."); return false; }
  },
  remove(key) {
    try { localStorage.removeItem(key); } catch (e) { /* ignore */ }
  },
  keys() {
    const out = [];
    try {
      for (let i = 0; i < localStorage.length; i++) {
        const k = localStorage.key(i);
        if (k && k.startsWith(PREFIX)) out.push(k);
      }
    } catch (e) { /* ignore */ }
    return out.sort();
  }
};

/* ----- State ----- */
let cart = {};
let orders = [];

function loadState() {
  const rawCart = store.get(KEYS.cart, {});
  cart = {};
  if (rawCart && typeof rawCart === "object" && !Array.isArray(rawCart)) {
    MENU.forEach(m => {
      const q = Number(rawCart[m.id]);
      if (Number.isInteger(q) && q > 0) cart[m.id] = q;
    });
  }
  const rawOrders = store.get(KEYS.orders, []);
  orders = Array.isArray(rawOrders) ? rawOrders.filter(o => o && Array.isArray(o.items)) : [];
}
```

```

/* ----- Utilities ----- */
const $ = id => document.getElementById(id);
const money = cents => "RM " + (cents / 100).toFixed(2);
const esc = s => String(s).replace(/[\<>'"]/g, c => ({ "&": "&amp;", "<": "&lt;", ">": "&gt;", "'": "&quot;", "\"": "&#39;" }[c]));
const menuItem = id => MENU.find(m => m.id === id);
let toastTimer;
function toast(msg) {
  const t = $("toast");
  t.textContent = msg;
  t.classList.add("show");
  clearTimeout(toastTimer);
  toastTimer = setTimeout(() => t.classList.remove("show"), 2200);
}

/* ----- Theme ----- */
function applyTheme(theme) {
  document.documentElement.dataset.theme = theme;
  $("themeBtn").textContent = theme === "dark" ? "☀️ Light mode" : "🌙 Dark mode";
}
$("themeBtn").addEventListener("click", () => {
  const next = document.documentElement.dataset.theme === "dark" ? "light" : "dark";
  applyTheme(next);
  store.set(KEYS.theme, next);
  renderStorage();
});

/* ----- Menu + table selectors ----- */
function renderMenu() {
  $("menu").innerHTML = MENU.map(m =>
    `<button type="button" data-id="${m.id}"><span class="icon">${m.icon}</span><span>${esc(m.name)}</span><span
class="price">${money(m.price)}</span></button>`
  ).join("");
}
function renderTableSelectors() {
  $("tableSel").innerHTML = TABLES.map(t => `<option>${esc(t)}</option>`).join("");
  const saved = store.get(KEYS.table, TABLES[0]);
  $("tableSel").value = TABLES.includes(saved) ? saved : TABLES[0];
  $("filterTable").innerHTML = ["All tables"].concat(TABLES).map(t => `<option>${esc(t)}</option>`).join("");
}
$("tableSel").addEventListener("change", e => { store.set(KEYS.table, e.target.value); renderStorage(); });
$("filterTable").addEventListener("change", renderHistory);

/* ----- Cart ----- */
function saveCart() {
  if (Object.keys(cart).length) store.set(KEYS.cart, cart);
  else store.remove(KEYS.cart);
}
function cartTotal() {
  return Object.entries(cart).reduce((sum, [id, q]) => sum + menuItem(id).price * q, 0);
}
function renderCart() {
  const ids = Object.keys(cart);
  $("cartEmpty").hidden = ids.length > 0;
  $("cartList").innerHTML = ids.map(id => {
    const m = menuItem(id), q = cart[id];
    return `<li><span class="name">${m.icon} ${esc(m.name)}</span>
      <span class="qty"><button type="button" data-act="dec" data-id="${id}" aria-label="Decrease">-</button>
<b>${q}</b><button type="button" data-act="inc" data-id="${id}" aria-label="Increase">+</button></span>
      <span>${money(m.price * q)}</span></li>`;
  }).join("");
  $("cartTotal").textContent = money(cartTotal());
}
$("menu").addEventListener("click", e => {
  const btn = e.target.closest("button[data-id]");
  if (!btn) return;
  cart[btn.dataset.id] = (cart[btn.dataset.id] || 0) + 1;
  saveCart(); renderCart(); renderStorage();
});
$("cartList").addEventListener("click", e => {
  const btn = e.target.closest("button[data-act]");
  if (!btn) return;
  const id = btn.dataset.id;
  cart[id] = (cart[id] || 0) + (btn.dataset.act === "inc" ? 1 : -1);
  if (cart[id] <= 0) delete cart[id];
  saveCart(); renderCart(); renderStorage();
});

```

```

$("clearCartBtn").addEventListener("click", () => {
  cart = {}; saveCart(); renderCart(); renderStorage();
});

/* ----- Orders ----- */
$("placeBtn").addEventListener("click", () => {
  if (!Object.keys(cart).length) { toast("Add at least one item first."); return; }
  const no = Number(store.get(KEYS.counter, 0)) + 1;
  const items = Object.entries(cart).map(([id, qty]) => {
    const m = menuItem(id);
    return { id, name: m.name, price: m.price, qty };
  });
  const order = { no, time: new Date().toISOString(), table: $("tableSel").value, items, total: cartTotal() };
  orders.unshift(order);
  if (!store.set(KEYS.orders, orders)) { orders.shift(); return; }
  store.set(KEYS.counter, no);
  cart = {}; saveCart();
  renderAll();
  toast("Order #" + no + " saved for " + order.table + ".");
});
$("history").addEventListener("click", e => {
  const btn = e.target.closest("button[data-del]");
  if (!btn) return;
  const no = Number(btn.dataset.del);
  if (!confirm("Delete order #" + no + "?")) return;
  orders = orders.filter(o => o.no !== no);
  store.set(KEYS.orders, orders);
  renderAll();
});
function renderHistory() {
  const filter = $("filterTable").value;
  const list = orders.filter(o => filter === "All tables" || o.table === filter);
  $("histEmpty").hidden = list.length > 0;
  $("history").innerHTML = list.map(o => {
    const d = new Date(o.time);
    return `<li>
      <div class="head"><span>Order #${o.no} · ${esc(o.table)}</span><span>${money(o.total)}</span></div>
      <div class="meta">${d.toLocaleDateString()} ${d.toLocaleTimeString([], { hour: "2-digit", minute: "2-digit"
    }}</div>
      <ul>${o.items.map(i => `<li>${i.qty} × ${esc(i.name)}</li>`).join("")}</ul>
      <div class="foot"><span></span><button class="del" type="button" data-del="${o.no}">Delete</button></div>
    </li>`;
  }).join("");
}

/* ----- Summary ----- */
function renderStats() {
  const today = new Date().toDateString();
  const todays = orders.filter(o => new Date(o.time).toDateString() === today);
  const sales = todays.reduce((s, o) => s + o.total, 0);
  const allSales = orders.reduce((s, o) => s + o.total, 0);
  const tally = {};
  todays.forEach(o => o.items.forEach(i => { tally[i.name] = (tally[i.name] || 0) + i.qty; }));
  const best = Object.entries(tally).sort((a, b) => b[1] - a[1])[0];
  $("stats").innerHTML = [
    ["Orders today", todays.length],
    ["Sales today", money(sales)],
    ["Best seller today", best ? `${esc(best[0])} (${best[1]})` : "-"],
    ["All-time sales", money(allSales)]
  ].map(([label, value]) => `<div class="stat"><small>${label}</small><strong>${value}</strong></div>`).join("");
}

/* ----- Storage inspector ----- */
function renderStorage() {
  const rows = store.keys().map(k => {
    let size = 0;
    try { size = k.length + (localStorage.getItem(k) || "").length; } catch (e) { /* ignore */ }
    return `<tr><td><code>${esc(k)}</code></td><td>${esc(KEY_INFO[k] || "Other")}</td><td>${size}</td></tr>`;
  });
  $("storageTable").querySelector("tbody").innerHTML =
    rows.length ? rows.join("") : `<tr><td colspan="3">Nothing stored yet.</td></tr>`;
}
$("exportBtn").addEventListener("click", () => {
  const data = { exportedAt: new Date().toISOString(), orders, currentOrder: cart };
  const blob = new Blob([JSON.stringify(data, null, 2)], { type: "application/json" });
  const a = document.createElement("a");
  a.href = URL.createObjectURL(blob);
  a.download = "mamak-orders.json";
}

```

```

document.body.appendChild(a); a.click(); a.remove();
setTimeout(() => URL.revokeObjectURL(a.href), 1000);
toast("Exported mamak-orders.json");
});
$("resetBtn").addEventListener("click", () => {
  if (!confirm("Delete ALL saved orders and settings for this app?")) return;
  store.keys().forEach(k => store.remove(k));
  loadState();
  applyTheme("light");
  renderTableSelectors();
  renderAll();
  toast("All data cleared.");
});

/* ----- Sync between tabs ----- */
window.addEventListener("storage", e => {
  if (e.key === null || (e.key && e.key.startsWith(PREFIX))) {
    loadState();
    applyTheme(store.get(KEYS.theme, "light"));
    renderAll();
  }
});

/* ----- Start ----- */
function renderAll() { renderCart(); renderHistory(); renderStats(); renderStorage(); }
applyTheme(store.get(KEYS.theme, "light"));
loadState();
renderMenu();
renderTableSelectors();
renderAll();

```